

Unequal Climate Policy in an Unequal World

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Motivation

Climate change and inequality

- ▶ Climate change is a critical challenge for mankind.
- ▶ Addressing climate change is difficult because there is **inequality**: across countries, across generations, **across households**.

Motivation

When does Inequality Matter?

- ▶ Climate change is an externality problem → Pigouvian tax!
- ▶ Inequality is easy to fix → Transfers!
- ▶ But what if policymakers cannot eliminate inequality?
 - ▶ political/institutional constraints
 - ▶ insufficient instruments to redistribute
- ▶ We focus on climate policy as a means to address the externality, not for redistribution.
- ▶ But, climate policy has redistributive effects (can potentially hurt the poor).

Related Literature

- ▶ **Carbon taxation with representative agent:** Nordhaus and Boyer (2003), Nordhaus (2007), Golosov et. al. (2014), Barrage (2018), Belfiori (2017).
- ▶ **Carbon taxation with heterogeneous agents:** Jacobs and Van Der Ploeg (2019), Douenne, Hummel and Pedroni (2023), Belfiori and Macera (2024), Fried et al. (2018, 2023), Krusell and Smith (2022), Känzig (2022), Bourany (2024).
- ▶ **Distributional role of carbon tax revenue:** Rausch et al. (2011), Pizer and Sexton (2019), Fullerton and Monti (2013), Goulder et. al. (2019).
- ▶ **Inequality and Carbon:** Sager (2019), Levinson and O'Brien (2019), Grainger and Kolstad (2010).

What we do

- ▶ This paper studies climate policy in an economy with heterogeneous households, clean and dirty consumption, and a climate externality from the dirty good. Three parts:
 - ▶ **DATA:** low-income households have higher carbon intensity per dollar spent (carbon tax is regressive).
 - ▶ **THEORY:** characterize optimal carbon tax rules under different constraints on the planner's problem.
 - ▶ **QUANTITATIVE:** we embed the simple model in a heterogeneous-agent climate-model calibrated to the US economy, and quantify the effects of taxes on the economy, climate, and welfare.

A Win-Win Climate Policy

- ▶ The benefits from a better climate (reduction in global temperature) more than compensate the disutility from the distorting consumption bundles
 - ▶ There are no income effects, by construction
- ▶ Win-win climate policy is possible under some optimal carbon tax formulas. Welfare gains in the aggregate but also for every individual.
 - ▶ Everyone (present and future, rich and poor) is better-off.

Roadmap

1. Use simple model to characterize the optimal carbon tax formula under various constraints
 - ▶ no constraints $\Rightarrow$ first-best Pigouvian tax
 - ▶ constrained efficiency: no transfers across households
 - ▶ constrained efficiency with uniform carbon tax across households
2. Generalize the analytical characterizations to a richer model
3. A quantitative application
 - ▶ Heterogeneous-agent macro-climate model
 - ▶ Calibration
 - ▶ Evaluate the aggregate and welfare consequences of enacting carbon taxes informed by theory

Simple Model

Model

- ▶ Economy populated by a continuum of households, indexed by i , with measure μ_i
- ▶ Two consumption goods, clean and dirty: (c_{ct}^i, c_{dt}^i)
- ▶ Production of the clean and the dirty good uses labor to produce according to: $Y_{ct} = N_{ct}$; $Y_{dt} = N_{dt}$
- ▶ Consumption of the dirty good adds carbon to the atmosphere:

$$S_{t+1} = S_t(1 - \delta) + v \sum_i \mu_i c_{dt}^i$$

Households

- ▶ Households' preferences over consumption and atmospheric carbon are given by

$$\sum_{t=0}^{\infty} \beta^t [u(c_{ct}^i, c_{dt}^i) - x(S_{t+1})]$$

where x is the climate damage function, $x' > 0$ and $x'' > 0$

- ▶ Households choose consumption (c_{ct}^i, c_{dt}^i) to maximize utility subject to budget constraints

$$c_{ct}^i + (1 + \tau_{dt}^i) c_{dt}^i \leq w_t \varepsilon^i + T_t^i, \quad \forall t$$

where ε^i is the labor endowment, inelastically supplied.

Optimal Climate Policy

- The optimal allocation (with weights α_i) is the solution to

$$\begin{aligned} \max_{\{c_{ct}^i, c_{dt}^i, S_{t+1}\}_t} \quad & \sum_i \alpha_i \sum_{t=0}^{\infty} \beta^t [u(c_{dt}^i, c_{ct}^i) - x(S_{t+1})] \\ \text{s.t.} \quad & S_{t+1} \geq (1 - \delta)S_t + v \sum_i \mu_i c_{dt}^i \quad \dots \quad (\beta^t \sigma_t) \\ & \sum_i \mu_i (c_{ct}^i + c_{dt}^i) \leq \sum_i \mu_i \varepsilon^i \quad \dots \quad (\beta^t \lambda_t) \end{aligned}$$

- The first order condition with respect to S_{t+1} implies:

$$\sigma_t = \underbrace{\sum_{j=1}^{\infty} [\beta(1 - \delta)]^{j-1} x'(S_{t+j})}_{\text{social marginal damages}}$$

Characterization of Optimal Allocation

- ▶ The planner wants to affect the relative price between clean and dirty consumption to reflect the social cost of carbon

$$\frac{u_{dt}^i}{u_{ct}^i} = 1 + \boxed{\frac{v\sigma_t^*}{u_{ct}^*}} \quad \begin{array}{l} \text{Social Marginal Damages} \\ \text{Social Value of Consumption} \end{array}$$

- ▶ The optimal allocation can be implemented in a market economy with

$$\tau_t^* = \frac{v\sigma_t^*}{u_{ct}^*} \quad \forall t$$

and non-uniform transfers

$$T_t^i = c_{ct}^i + (1 + \tau_t^*)c_{dt}^i - \varepsilon^i$$

Redistribution in the Optimal Policy

- ▶ If the planner is Utilitarian (i.e. $\alpha_i = \mu_i$), the optimal allocation involves significant redistribution across households.
- ▶ What if the planner uses Negishi weights (i.e. no incentive to redistribute)?
 - ▶ under certain conditions, we can show that the Negishi optimal carbon tax is greater than the Utilitarian optimal tax.
- ▶ In what follows, we focus on a Utilitarian planner with constrained efficiency: **no net transfers across households** (à la Davila, Hong, Krussel, Rios-Rull 2012)

Constrained Efficient Climate Policy

- The optimal allocation $\{c_{ct}^i, c_{dt}^i, S_{t+1}\}_{t=0}^{\infty}$ maximizes

$$\sum_i \mu_i \sum_{t=0}^{\infty} \beta^t [u(c_{dt}^i, c_{ct}^i) - x(S_{t+1})] \quad \text{s.t.}$$

$$S_{t+1} \geq (1 - \delta)S_t + v \sum_i \mu_i c_{dt}^i$$

$$c_{dt}^i + c_{ct}^i \leq \varepsilon^i \quad \forall i, t$$

Proposition [Constrained Efficient Climate Policy]

Let $\{\tau_t^i, T_t^i\}_{t=0, \forall i}^\infty$ be a sequence of carbon taxes on dirty consumption and lump-sum transfers. The constrained efficient allocation can be implemented in a market economy with:

$$\tau_t^i = \frac{v\sigma_t}{u_{ct}^i}; \quad T_t^i = \tau_t^i c_{dt}^i \quad \forall i, t$$

1. The transfers guarantee that the consumer's budget constraint holds for all i .
2. The constrained efficient carbon tax is similar to the optimal one, but deviates in an important way: τ_t^i higher for wealthier households because they have a lower marginal utility

Takeaway

- ▶ The constrained efficient carbon tax in a heterogeneous economy is heterogeneous (and progressive)

Can we make it homogeneous?

- ▶ To get a homogeneous carbon tax as the outcome of an optimal policy design for a heterogeneous-agent economy, we must impose uniformity of the tax rate as an additional constraint in the planning problem.
- ▶ Using the optimality conditions, we must impose:

$$\frac{u_{dt}^i}{u_{ct}^i} = \frac{u_{dt}^j}{u_{ct}^j} \quad \forall i, j, t$$

Uniform Constrained-Efficient Climate Policy

- The uniform constrained-efficient allocation $\{c_{ct}^i, c_{dt}^i, S_{t+1}\}_{t=0}^{\infty}$ maximizes

$$\sum_i \mu_i \sum_{t=0}^{\infty} \beta^t [u(c_{dt}^i, c_{ct}^i) - x(S_{t+1})] \quad \text{s.t.}$$

$$S_{t+1} \geq (1 - \delta)S_t + v \sum_i \mu_i c_{dt}^i$$

$$c_{dt}^i + c_{ct}^i \leq \varepsilon^i \quad \forall i, t$$

$$\frac{u_{dt}^i}{u_{ct}^i} = \frac{u_{dt}^j}{u_{ct}^j} \quad \forall i, j, t$$

Proposition [Uniform Constrained-Efficient Carbon Tax]

Let $\{\tau_t, T_t^i\}_{t=0, \forall i}^\infty$ be a sequence of carbon taxes on dirty consumption and lump-sum transfers. The uniform constrained efficient allocation can be implemented in a market economy with:

$$\tau_t = \frac{v\sigma_t}{\sum_i \frac{\mu_i c_t^i}{\sum_j \mu_j c_t^j} u_{ct}^i}; \quad T_t^i = \tau_t c_{dt}^i \quad \forall i, t$$

- ▶ The uniform constrained-efficient carbon tax uses a weighted average of marginal utilities to price the climate externality.
- ▶ When risk aversion is greater than 1, consumption-weighted average marginal utility is higher than marginal utility of average consumption $\Rightarrow$ higher social value of consumption

Corollary [Alternative Implementation]

Let $\{\tau_{dt}, \tau_{ct}, T_t\}_{t=0}^{\infty}$ be a sequence of **uniform** carbon taxes on dirty consumption, **uniform** clean subsidy on clean consumption, and a **uniform** lump-sum transfer. The uniform constrained efficient allocation can be implemented in a market economy with:

$$\tau_{dt} = \gamma \tau_t, \quad \tau_{ct} = (1 - \gamma) \frac{\tau_t}{1 + \tau_t}, \quad T_t = \tau_{ct} \bar{c}, \quad \forall t$$

- This alternative implementation is more feasible as all of the policy instruments are uniform, i.e. not individual specific.

Quantifying the welfare gains

- ▶ Heterogeneous carbon tax delivers highest average initial welfare gain and largest long-run temperature reduction
- ▶ Constrained efficient uniform carbon tax leads to a smaller temperature reduction but is Pareto improving
- ▶ Negishi optimal carbon tax has lowest support, hurting poor households

Policy	Avg. welfare gain (%)	Support	LR temp. reduction
Heterogeneous tax w/ rebate	0.068	90.0	1.9
Uniform tax with rebate	0.063	100.0	1.2
Negishi optimal tax w/ rebate	0.060	76.0	1.7

Generalizations of the Optimal Tax Formulas

- ▶ In the paper, we generalize the optimal tax derivations to include
 - ▶ non-homothetic preferences
 - ▶ stochastic shocks
 - ▶ endogenous labor and capital
 - ▶ ad hoc borrowing constraints
- ▶ We now turn to quantitative analysis that incorporate these features to evaluate the aggregate and welfare consequences of implementing climate policies informed by our theory.

Quantitative Model

Model

- ▶ Heterogeneous agents model with incomplete markets (Aiyagari, Bewley, Hugget, Imrohoroglu)
- ▶ Economy populated by a continuum of households, indexed by i , with measure μ_i
- ▶ Two consumption goods, clean and dirty: (c_{ct}^i, c_{dt}^i)
- ▶ Production of the clean and the dirty good uses labor and capital with CRS technology: $Y_{ct} = F(K_{ct}, N_{ct})$;
 $Y_{dt} = F(K_{dt}, N_{dt})$
- ▶ Consumption of the dirty good adds carbon to the atmosphere:

$$S_{t+1} = S_t(1 - \delta) + v \sum_i \mu_i c_{dt}^i \quad (1)$$

Households

- Preferences over consumption, **leisure**, and carbon:

$$\mathbf{E}_0 \sum_{t=0}^{\infty} \beta^t [u(c_{ct}^i, c_{dt}^i, \ell_t^i) - x(S_{t+1})]$$

- Households choose consumption (c_{ct}^i, c_{dt}^i) , **capital** k_t^i , and **labor** n_t^i to maximize utility subject to

$$\begin{aligned} p_t c_{dt}^i (1 + \tau_t) + c_{ct}^i + k_{t+1}^i - k_t^i &\leq y_t^i, \quad \forall t \\ k_{t+1}^i &\geq 0, \quad \forall t \end{aligned}$$

where $\ell_t^i = 1 - n_t^i$, and

$$y_t^i \equiv w_t \varepsilon_t^i n_t^i - T(w_t \varepsilon_t^i n_t^i) + (r_t - \delta_k) k_t^i (1 - \tau_{kt}) + Tr_t^i$$

Calibration strategy

- ▶ Many model parameters affect both economic and climate outcomes (e.g., discount factor, disutility of work, preference for dirty good)
- ▶ Other parameters affect climate only (e.g., disutility from climate damage, carbon content of dirty consumption, carbon absorption). Why?
 1. climate damage enters separably in utility
 2. carbon does not directly affect production
- ▶ This allows us to first calibrate the model's economic parameters to an “economic” steady state, and then calibrate the model's climate parameters

Calibration: Preferences

- Utility function:

$$u(c_c, c_d, \ell) = \frac{((c_c + \bar{c})^\gamma c_d^{1-\gamma})^{1-\kappa}}{1-\kappa} - \phi \frac{(1-\ell)^{1+\nu}}{1+\nu}$$

Parameters	Values	Targets / Source
Discount factor β	0.97	Wealth-to-GDP: 4.8 (2014)
Risk aversion κ	2	Standard value
Labor disutility, ϕ	29.6	Average hours: 30 percent
Frisch elasticity $1/\nu$	0.5	Standard value
Clean share γ	0.97	\$50/ton carbon tax leads to 0.8 degree reduction from BAU
Non-homotheticity $\bar{c}$	0.16	emissions intensity 31% higher for  low-income than high-income households

Calibration: Technology and Shocks

- ▶ Production function: $F(K, N) = K^\alpha N^{1-\alpha}$
- ▶ Normal productivity: $\log(\varepsilon_t^i) = \rho \log(\varepsilon_{t-1}^i) + \xi_t^i$, $\xi_t^i \sim N(0, \sigma_\varepsilon^2)$
- ▶ Superstar state ε_{sup} to match wealth/earnings distribution

Parameters	Values	Targets / Source
Capital weight, α	0.36	capital income share: 36%
Capital depreciation, δ_k	0.05	standard value
Productivity persistence ρ	0.94	author estimates
Standard deviation, σ_ε	0.20	earnings Gini: 0.47
Superstar parameters		
productivity, $\varepsilon_{sup}/\varepsilon_{med}$	163	wealth share top 1.0%: 34%
persistence, $\pi(\varepsilon_{sup}, \varepsilon'_{sup})$	0.94	wealth Gini: 0.83
entry probability, $\pi(1 : 9, \varepsilon'_{sup})$	6e-5	fraction of superstars: 0.1%

Calibration: Government

- Progressive earnings tax (Benabou, HSV,...)

$$T(y) = y - \tilde{y}^{\nu_y} \frac{1 - \tau_y}{1 - \nu_y} y^{1 - \nu_y}$$

where $\tilde{y}^{\nu_y}$ is average earnings

Parameters	Values	Targets / Source
Average tax parameter, τ_y	0.23	average labor income tax: 13%
Progressivity parameter, ν_y	0.17	37.9% marginal tax rate on top 1% earners
Capital income tax, τ_k	0.27	Carey and Rabesona (2002)
Consumption tax, τ_c	0.06	Carey and Rabesona (2002)

Calibration: Climate

- ▶ Temperature function: $T_t = \frac{\lambda}{\log(2)} \log \left(\frac{S_t}{\bar{S}} \right)$ (Golosov et al. 2014)
- ▶ Climate damage function: $x(S) = \frac{\psi}{2} S^2$

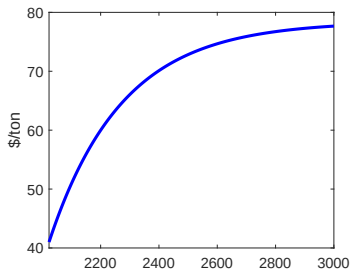
Parameters	Values	Targets / Source
Carbon absorption, δ	1/300	average life of carbon: 300 years
Carbon intensity, v	326.4	1.4 degree increase by 2100 under BAU
Climate disutility, ψ	0.04	welfare loss from 2.5 degree increase ≈ 1.74 percent output reduction
Temperature parameters		
climate sensitivity, λ	3	doubling of carbon $\Rightarrow$ 3-degree increase
initial carbon, $\bar{S}$	581	pre-industrial carbon stock (gigatons)

Quantitative Exercises

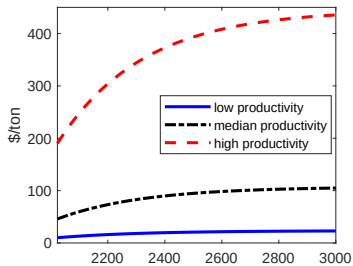
Carbon tax

- ▶ We use the Pigouvian tax formulas from our propositions
- ▶ Tax formulas depend on endogenous variables
 - ▶ carbon stocks, marginal utilities
 - ▶ start with BAU allocations to calculate carbon tax; solve transition;
 - ▶ update carbon tax with new allocations; repeat until convergence
- ▶ For today, heterogeneous carbon tax based on current productivity (+ history in progress)

(a) Uniform carbon tax

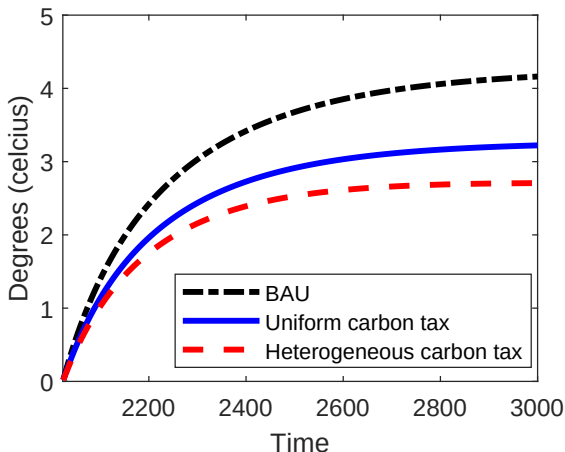


(b) Heterogeneous carbon tax



Global Temperature

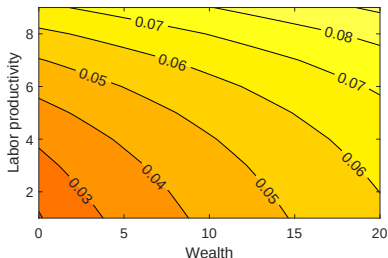
- The carbon tax leads to a 1–2 degree reduction compared to BAU



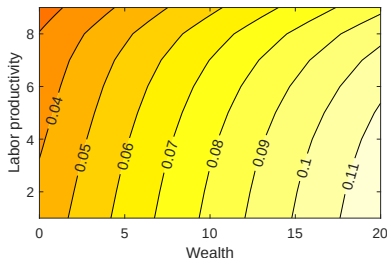
Initial distribution of welfare

- ▶ Welfare gains (relative to BAU) positive for virtually all households
- ▶ Pareto improving in the uniform case

(a) Uniform carbon tax



(b) Heterogeneous carbon tax

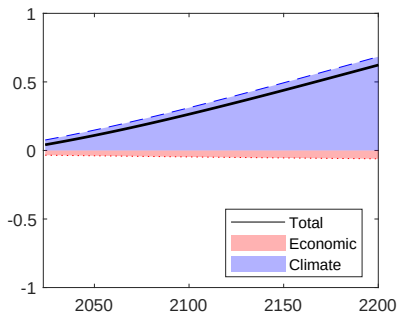


Units: Permanent consumption equivalents (percent)

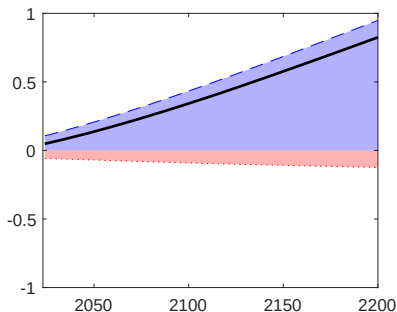
Average welfare decomposition over time

- Average welfare gains become large over time

(a) Homogeneous carbon tax



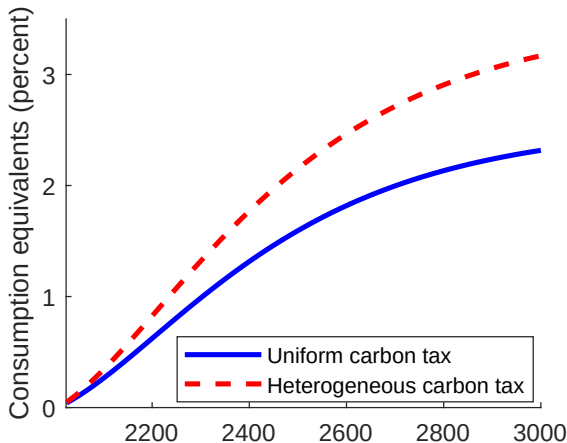
(b) Heterogeneous carbon tax



Units: Permanent consumption equivalents (percent)

Average welfare gains over time

- Average welfare gains larger for the heterogeneous carbon tax, both initially and throughout the transition



Alternative exercises

- ▶ A more **implementable policy** with uniform carbon tax, uniform clean subsidy, and uniform lumpsum transfer delivers climate and welfare gains similar to the individual rebate.
- ▶ A **constant carbon tax** that delivers a similar level of carbon reduction has much lower welfare gains and support.

Policy	Avg. welfare gain (%)	Support	LR temp. reduction
Heterogeneous tax with rebate	0.052	99.9	1.6
Uniform tax with rebate	0.045	100.0	1.0
Uniform tax with subsidy/transfer	0.047	100.0	1.0
Constant tax (\$100/ton) w/ rebate	0.009	49.7	1.2

Conclusion

- ▶ We derive optimal carbon tax rules for economies with inequality.
- ▶ Well-designed (informed by theory) climate policies can lead to Pareto improving welfare gains.
- ▶ Next steps:
 - ▶ Heterogeneous climate damages (if low-income households are more vulnerable to climate damages, there is more room for a Pareto-improvement)
 - ▶ Implementation of the constrained-efficient allocation with climate targets.

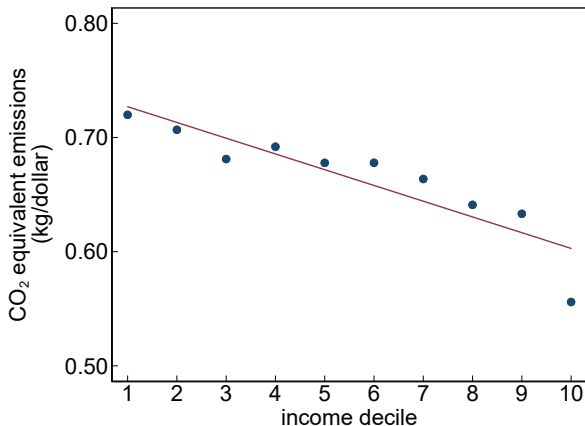
Appendix

Data

[back to calibration](#)

- ▶ We combine two datasets
- ▶ Environmental Protection Agency (EPA)
 - ▶ embodied emissions for 460 commodities
(covering cradle → factory gate → shelf) [details](#)
- ▶ Consumer Expenditure Survey (CEX, 2019)
 - ▶ 671 expenditure categories
 - ▶ 5000+ working-age households
 - ▶ construct CEX-NAICS concordance [examples](#)
- ▶ Compute CO₂-equivalent embodied emissions per dollar spent, for each household

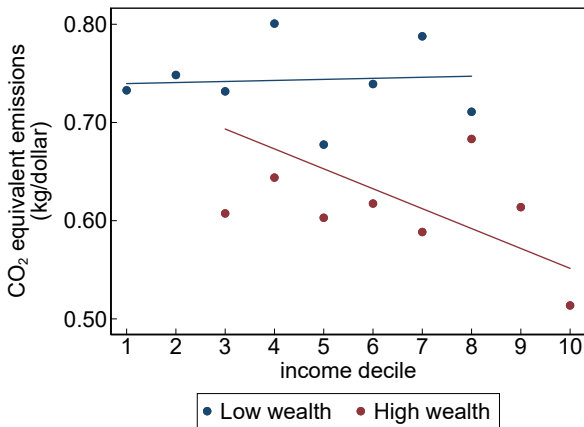
Embodied emission shares higher for low-income



[back to calibration](#)

[PSID](#)

Embodied emission shares higher for low-income and low-wealth



Embodied emissions and expenditure shares

- Emission intensities are higher for low-income households because they spend relatively more on high-intensity goods

Expenditure category	Embodied emissions (CO ₂ kg/dollar)	Expenditure shares (percent)	
		Low income	High income
Utilities	1.71	11.0	6.8
Transportation	1.09	22.3	21.4
Food/Beverages at home	0.80	17.1	10.2
Household furnishings/equipment	0.28	2.5	4.4
Food/Beverages away from home	0.21	5.7	8.4
Clothing and footwear	0.20	2.3	3.5
Education and child care	0.18	1.0	9.3
Entertainment	0.15	4.0	7.2
Health care	0.14	7.2	9.5
Shelter	0.11	21.4	11.8
Other expenditures	0.10	5.6	7.6

High and low income correspond to the top and bottom deciles of income, respectively, conditional on working age. [back to calibration](#)

Summary of empirical findings

[back to calibration](#)

- ▶ Embodied emission intensities decline with income and wealth
- ▶ Robust to controlling for household characteristics: [Regressions](#)
 - ▶ household head age
 - ▶ household head education
 - ▶ household size
- ▶ Robust to:
 - ▶ alternative emissions dataset (FRS)
 - ▶ alternative expenditure dataset (PSID)

Selected examples of UCC-Naics concordance [back](#)

UCC	Description	NAICS	Description	CO ₂ e emissions (kg/2018 USD)
100210	Cheese	311513	Cheese Manufacturing	1.585
90110	Fresh Milk All Types	311511	Fluid Milk Manufacturing	1.323
80110	Eggs	112300	Chicken Egg Production	1.052
140110	Frozen Vegetables	311411	Frozen Fruit, Juice, Vegetable Mfg.	.846
610310	Pet Food	311111	Dog and Cat Food Mfg.	.75
530210	Intercity bus fares	485210	Interurban/Rural Bus Transportation	.515
170110	Cola Drinks	312111	Soft Drink Manufacturing	.444
190212	Dinner At Full Service	722511	Full-Service Restaurants	.255
450220	New Motorcycles	336991	Motorcycle, Bicycle, and Parts Mfg.	.254
370314	Boys pants and shorts	315220	Men's/Boys' Cut/Sew Apparel Mfg.	.187
630110	Cigarettes	312230	Tobacco Manufacturing	.153
560110	Physicians Services	621111	Offices of Physicians	.082

Details on embodied emissions data (EPA) [back](#)

- ▶ Included greenhouse gases: CO₂, Methane (CH₄), Nitrous Oxide (N₂), Other GHGs
- ▶ Convert to CO₂ using IPCC (The Intergovernmental Panel on Climate Change) AR4 (Assessment Report) GWP-100 (Global warming potential over 100 years, compared to CO₂)
- ▶ covers supply chain emissions (cradle to factory gate) and also margins (factory gate to shelf, including transportation, wholesale and retail)
- ▶ Environmentally-Extended Input-Output (EEIO) model
 - ▶ compute direct requirement matrix using Make/Use Tables
 - ▶ compute total requirement matrix using Leontief Inverse
 - ▶ combine with direct emissions factors from National Greenhouse Gas Industry Attribution Model (NGIAM)

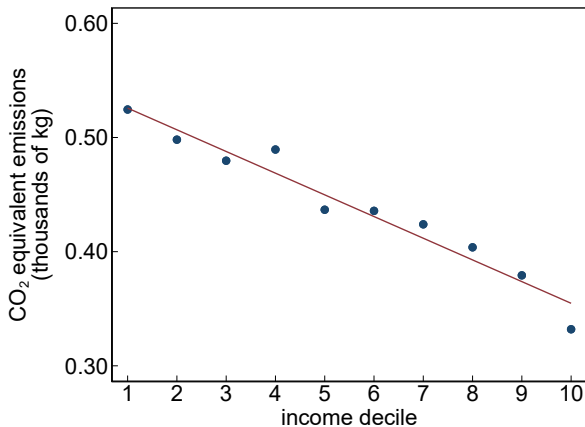
Embodied emission shares [back](#)

	(1)	(2)	(3)	(4)
Wealth	-1.68*** (0.053)		-1.35*** (0.068)	-1.10*** (0.070)
Income		-3.69*** (0.095)	-1.79*** (0.224)	-2.83*** (0.247)
College=1				-4.51*** (0.381)
Observations	16368	56122	16368	16368
Adjusted R^2	0.057	0.026	0.060	0.135

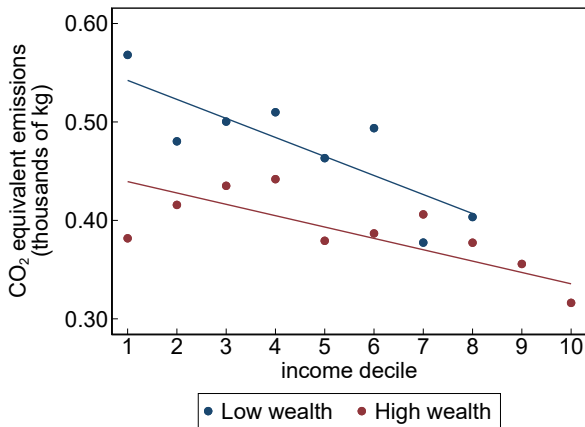
Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$
(4) additionally includes age and family size fixed effects.

- ▶ $\text{sd}(\log(\text{wealth})) = 3.3$
- ▶ $\text{sd}(\log(\text{income})) = 1.0$

Emission shares higher for low-income (PSID) [back](#)



Embodied emission shares higher for low-income and low-wealth (PSID) [back](#)



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	(1)	(2)	(3)	(4)
Wealth	-1.68*** (0.053)		-1.35*** (0.068)	-1.10*** (0.070)
Income		-3.69*** (0.095)	-1.79*** (0.224)	-2.83*** (0.247)
College=1				-4.51*** (0.381)
Observations	16368	56122	16368	16368
Adjusted R^2	0.057	0.026	0.060	0.135

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$
(4) additionally includes age and family size fixed effects.

- ▶ $\text{sd}(\log(\text{wealth})) = 3.3$
- ▶ $\text{sd}(\log(\text{income})) = 1.0$